

MHT Package — Stata Documentation

Version 2.0.0 | Viviano, Wuthrich, and Niehaus (2026)

"A Model of Multiple Hypothesis Testing" — *arXiv:2104.13367v10*

1. Overview

The `mht` package implements the **economically optimal multiple hypothesis testing (MHT)** framework from Viviano, Wuthrich, and Niehaus (2026). The optimal per-test significance level is (Proposition 4.1):

$$\alpha^*(J) = C(J, \text{Sigma}) / (b * \omega_bar(J))$$

This simplifies to closed-form expressions depending on the cost structure (Equations 26–27).

Commands

Command	Purpose
<code>mht_critical</code>	Compute optimal critical values (core formula)
<code>mht_test</code>	Test a variable of p-values with 5 procedures
<code>mht_est</code>	Postestimation: test J treatment coefficients
<code>mht_cost_estimate</code>	Estimate cost function parameters from data
<code>mht_table</code>	Generate reference tables of critical values

2. Installation

```
adopath + "path/to/critique/stata"
```

3. `mht_est` — Postestimation

Works after any standard estimation command (`regress`, `ivregress`, `logit`, `areg`, `xtreg`, `reghdfe`, etc.).

```
mht_est, vars(varlist) alphabar(#)  
      [model(string) cfshare(#) jbar(#) nmratio(#) mbar(#) beta(#) iota(#)  
      onesided twosided]
```

Options

Option	Type	Default	Description
<code>vars(varlist)</code>	varlist	—	Names of coefficients to test (must match <code>e(b)</code>)
<code>alphabar(#)</code>	(0,1)	—	Benchmark significance level
<code>onesided / twosided</code>	flag	<code>onesided</code>	Direction of test; see Section 7
<code>mbar(#)</code>	numeric	—	Benchmark per-arm sample size. Sets <code>nm_ratio = (e(N)/J) / mbar</code> . Overrides <code>nmratio</code> .
<code>model(string)</code>	string	<code>"linear"</code>	<code>"linear"</code> or <code>"cobbdouglas"</code>
<code>cfshare(#)</code>	(0,1)	0.46	Fixed cost share (Linear model)
<code>jbar(#)</code>	> 0	3	Average arms per trial (Linear model)
<code>nmratio(#)</code>	> 0	1.0	Study sample size / benchmark
<code>beta(#)</code>	real	0.13	Cost elasticity w.r.t. arms (Cobb-Douglas)
<code>iota(#)</code>	real	0.075	Cost elasticity w.r.t. sample size (Cobb-Douglas)

Stored results

Scalars: `r(alpha_opt)`, `r(alpha_bonf)`, `r(alpha_bar)`, `r(J)`, `r(nm_ratio)`, `r(n_reject_opt)`, `r(n_reject_bonf)`, `r(n_reject_holm)`, `r(n_reject_bh)`, `r(n_reject_unadj)`, `r(coef_<var>)`, `r(se_<var>)`, `r(t_<var>)`, `r(p_<var>)`, `r(rej_opt_<var>)`, `r(rej_bonf_<var>)`, `r(rej_holm_<var>)`, `r(rej_bh_<var>)` for each tested variable.

Macros: `r(model)`, `r(vars)`.

Examples

```
* Basic usage
regress consumption treat1 treat2 treat3 baseline controls, cluster(hh_id)
mht_est, vars(treat1 treat2 treat3) alphabar(0.05)

* Cobb-Douglas calibration
mht_est, vars(treat1 treat2 treat3) alphabar(0.05) ///
    model(cobbdouglas) beta(0.13) iota(0.075)

* With benchmark sample size
mht_est, vars(treat1 treat2 treat3) alphabar(0.05) mbar(500)

* After areg with FE
areg consumption treat baseline controls, absorb(village) cluster(rand_unit)
mht_est, vars(treat) alphabar(0.05)

* Two-sided
logit adoption treat1 treat2, robust
mht_est, vars(treat1 treat2) alphabar(0.05) twosided
```

4. `mht_critical` — Compute Optimal Critical Values

Computes the optimal per-test significance level for a given number of hypotheses and cost model. Called internally by `mht_est`.

```
mht_critical, jhypotheses(#) alphabar(#)  
                [model(string) cfshare(#) jbar(#) nmratio(#) beta(#) iota#)]
```

Options

Option	Type	Default	Description
<code>jhypotheses(#)</code>	Integer ≥ 1	—	Number of hypotheses
<code>alphabar(#)</code>	(0,1)	—	Benchmark significance level
<code>model(string)</code>	string	"linear"	"linear" (Eq. 26) or "cobbdouglas"
<code>cfshare(#)</code>	(0,1)	0.46	Fixed cost share
<code>jbar(#)</code>	> 0	3	Average arms per trial
<code>nmratio(#)</code>	> 0	1.0	Study sample size / benchmark
<code>beta(#)</code>	real	0.13	Cost elasticity w.r.t. arms
<code>iota(#)</code>	real	0.075	Cost elasticity w.r.t. sample size

Stored results

Scalars: `r(alpha_opt)`, `r(t_star)`, `r(alpha_bonf)`, `r(t_bonf)`.

Examples

```
mht_critical, jhypotheses(5) alphabar(0.05)  
display "Optimal alpha: " r(alpha_opt) " Bonferroni: " r(alpha_bonf)  
  
mht_critical, jhypotheses(5) alphabar(0.05) model(cobbdouglas)
```

5. `mht_test` — Test a Variable of P-Values

Applies MHT adjustment to a pre-computed variable of p-values already in memory.

```
mht_test varname [if] [in], alphabar(#)  
                [model(string) zstat generate(string) replace  
                cfshare(#) jbar(#) nmratio(#) beta(#) iota#)]
```

Options

Option	Description
<code>varname</code>	Variable with one-sided p-values (or z-stats with <code>zstat</code>)
<code>alphabar(#)</code>	Benchmark significance level
<code>zstat</code>	Treat input as z-statistics
<code>generate(prefix)</code>	Prefix for output variables (default: <code>mht</code>)
<code>replace</code>	Replace existing output variables

Generated variables

`{prefix}_reject_opt`, `{prefix}_reject_bonf`, `{prefix}_reject_holm`, `{prefix}_reject_bh`, `{prefix}_reject_unadj`, `{prefix}_alpha_opt`.

6. `mht_cost_estimate` — Estimate Cost Function Parameters

Estimates research cost function from data on project costs, arms, and sample sizes.

```
mht_cost_estimate costvar armsvar sizevar, alphabar(#)
                    [model(string) controls(varlist) robust cluster(varname) table]
```

Models

Cobb-Douglas (default): $\log(C) = \text{const} + \beta * \log(J) + \iota * \log(n) + \text{controls}$

Linear: Estimates $C = c_f + c_v * J * n$ to recover $cf_share = c_f / \text{mean}(C)$.

Stored results

Scalars: `e(beta)`, `e(iota)`, `e(se_beta)`, `e(se_iota)`, `e(p_beta_0)`, `e(p_beta_1)`, `e(cf_share)` (Linear), `e(N)`.

Example

```
mht_cost_estimate proj_cost n_arms n_obs, alphabar(0.05) robust table
local beta_est = e(beta)
local iota_est = e(iota)

regress outcome treat1 treat2, cluster(hh_id)
mht_est, vars(treat1 treat2) alphabar(0.05) ///
        model(cobbdouglas) beta(`beta_est') iota(`iota_est')
```

7. `mht_table` — Reference Table of Critical Values

```
mht_table [, alphabar(#) jrange(numlist) nmratios(numlist) model(string)
          cfshare(#) jbar(#) beta(#) iota(#)]
```

Examples

```
mht_table                                * Table 1 (Linear, alpha_bar=0.05)
mht_table, model(cobbdouglas) jrange(1 3 5 9) nmratios(0.5 1.0 2.0)
mht_table, alphabar(0.10) jrange(1 2 3 4 5)
```

8. One-Sided vs. Two-Sided Tests

All commands default to **one-sided tests** (positive direction). This follows from the paper's model: the status quo remains when no positive finding is reported (Remark 4.1).

Use `twosided` when there is uncertainty about the policymaker's default action (Appendix B.1).

Setting	Correct test
Drug approval, scale-up	<code>onesided</code> (default)
Policymaker may act without recommendation	<code>twosided</code>

One-sided p-value convention: $t \geq 0$ gives $p = p_{\text{two}}/2$; $t < 0$ gives $p = 1 - p_{\text{two}}/2$ (> 0.5 , never rejected).

9. Interpreting Results

```
-----
MHT Postestimation Results
Viviano, Wuthrich, and Niehaus (2026)
After: regress
-----

Hypotheses tested:      3
Benchmark alpha:       0.0500
Cost model:             Linear (Eq. 26)
P-values:               one-sided (positive direction)
-----

Procedure              Test size    Rejections
-----
Optimal (model-based)  0.028571      2
Bonferroni             0.016667      1
Holm (step-down)       step-wise     2
BH (FDR control)       step-wise     2
```

Unadjusted	0.050000	3

The optimal α^* lies between Bonferroni and unadjusted. More fixed costs => closer to Bonferroni; more variable costs => closer to unadjusted.

10. Parameter Reference

Linear Model

Parameter	Option	Default	Source
Fixed cost share	<code>cfshare(#)</code>	0.46	Sertkaya et al. (2016)
Avg. arms per trial	<code>jbar(#)</code>	3	Pocock et al. (2002)

Cobb-Douglas (J-PAL)

Parameter	Option	Default	Source
Cost elasticity (arms)	<code>beta(#)</code>	0.13	Paper, Appendix A
Cost elasticity (sample)	<code>iota(#)</code>	0.075	Paper, Appendix A

beta = 0 => Bonferroni optimal; **beta = 1** => no adjustment; **beta = 0.13** => intermediate.

11. Complete Workflow

```
adopath + "path/to/critique/stata"

* Step 1: Reference table
mht_table, alphabar(0.05) jrange(1 2 3 5 10)

* Step 2: Regression with multiple treatments
use hh_data.dta, clear
local arms "treat_cash treat_inkind treat_training"
areg consumption `arms' baseline_consumption controls, ///
    absorb(village) cluster(hh_id)
mht_est, vars(`arms') alphabar(0.05)

* Step 3: Data-driven calibration (if cost data available)
use jpal_costs.dta, clear
mht_cost_estimate total_cost n_arms sample_size, alphabar(0.05) robust table
local est_beta = e(beta)
local est_iota = e(iota)

use hh_data.dta, clear
areg consumption `arms' baseline_consumption controls, ///
    absorb(village) cluster(hh_id)
```

```
mht_est, vars(`arms') alphabar(0.05) model(cobbdouglas) ///  
      beta(`est_beta') iota(`est_iota')
```

12. References

Viviano, D., K. Wuthrich, and P. Niehaus (2026). A model of multiple hypothesis testing. *arXiv:2104.13367v10*.

Sertkaya, A., H. H. Wong, A. Jessup, and T. Beleche (2016). Key cost drivers of pharmaceutical clinical trials in the United States. *Clinical Trials*, 13(2), 117-126.